AI Lab

Predicting Shapes using Random Forest in RapidMiner

1. Dataset

The dataset has 16 rows and four columns: Color, Outline, Dot and shape (the label). Shape is known for 14 rows; the last 2 rows (green, dashed, yes and yellow, solid, no) are to be predicted.

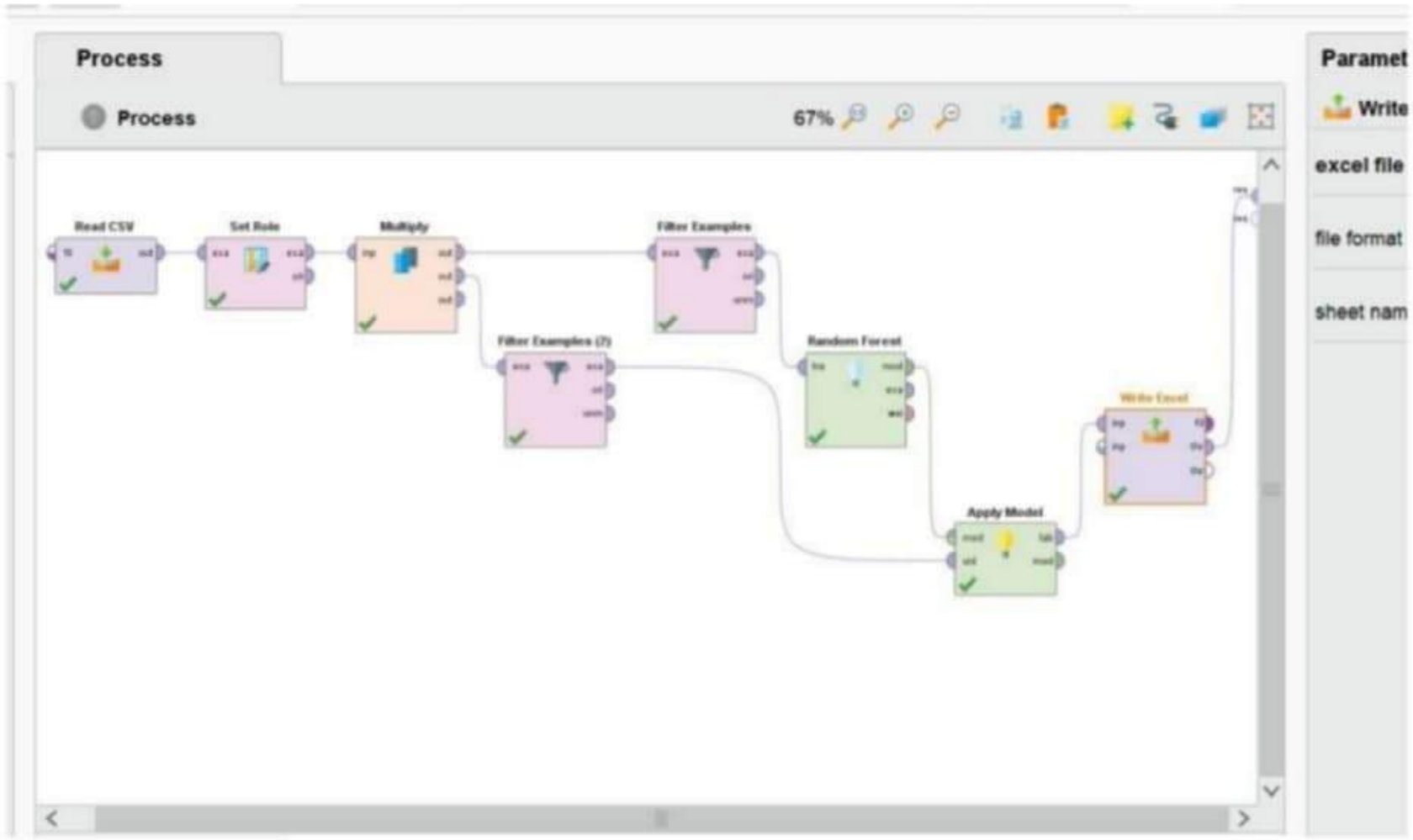
Color	Outline	Dot	shape
green	dashed	no	triangle
green	dashed	yes	triangle
yellow	dashed	no	square
red	dashed	no	square
red	solid	no	square
red	solid	yes	triangle
green	solid	no	square
green	dashed	no	triangle
yellow	solid	yes	square
red	solid	no	square
green	solid	yes	square
yellow	dashed	yes	square
yellow	solid	no	square
red	dashed	yes	triangle
green	dashed	yes	
yellow	solid	no	

Figure 1: Dataset with 14 known rows and 2 rows to be predicted

2. Process

Read CSV, Set Role (label = shape), Multiply, Filter Examples (known rows) into Random Forest, Filter Examples (2) (unknown rows), Apply Model, and Write Excel.

CamScanner



3. Result

Apply Model predicts the shape of both rows with a confidence for each class (triangle and square); the two values in each row add up to 1.

ExampleSet (Apply Model)

Open in Turbo Prep Auto Model Interactive Analysis

Row No.	shape	prediction(s...	confidence(...	confidence(...	Color	Outline	Dot
1	?	Triangle	0.810	0.190	green	dashed	yes
2	?	square	0.000	1.000	Yellow	solid	no

4. Updated Excel Sheet

The predicted rows are written to Excel with the original attributes, both class confidences and the prediction(shape) column.

Color	Outline	Dot	shape	confidence(triangle)	confidence(square)	prediction(shape)
green	dashed	yes		0.810	0.190	triangle
yellow	solid	no		0.000	1.000	square

5. Conclusion

Row 1 (green, dashed, yes): predicted shape is triangle, with a confidence of 81.0%.

Row 2 (yellow, solid, no): predicted shape is square, with a confidence of 100%.

CamScanner